① Program A:

8-Bit Addition and Subtraction:

Given: A = 5AH, B = 27H

Addition:

5AH + 27H = 81H

Sum = 81H, CY = 0

AC = 1, P = 1, Z = 0, S = 1

Flag Byte = 94H

Signed overflow = yes.

Subtraction:

5AH - 27H = 33H

Difference = 33H, Borrow = 0

AC = 0, P = 1, Z = 0, S = 0

Flag Byte = 06H

Signed overflow = NO

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② Program B: 16-bit Addition and Subtraction

Addition:

1234H + 4321H

= 5555H

Carry = 0

Subtraction:

1234H - 4321H

= CF13H

Borrow = 1.

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③ Program C: 8-Bit Multiplication and Division

Multiplication:

14H × 06H

= 0078H

Product = 0078H.

Division:

F2H ÷ 0AH

Quotient = 18H

Remainder = 02H.

Verification:

F2H = (18H × 0AH) + 02H.

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④ Program D : 16-bit Multiplication and Division

Multiplication:

1357H × 0028H

= 0002CA13H

Upper Word = 0003H

Lower Word = CA13H.

Division:

9A23H ÷ 0011H

Quotient = 0911H

Remainder = 0002H

Verification:

9A23H = (0911H × 0011H) + 0002H.